Weight perturbation across four AI weather models

Multiplicative weight noise $\tilde{W}_m = W \odot (1 + \sigma \xi_m)$, $\xi_m \sim \mathcal{N}(0,1)$, independent across members.

Phase 1: perturb all weights, sweep magnitude $\sigma \in \{0.001, 0.003, 0.01, 0.03, 0.1\}$. Phase 2: one layer group, $\sigma_{\text{group}} = \sigma_{\text{full}} \sqrt{N_{\text{total}}/N_{\text{group}}}$ (shown here at $\sigma_{\text{full}} = 0.01$).